

RESEARCH INTERESTS

With 4+ years of experience in intermittent computation, embedded systems, and hardware security, my research focuses on designing, evaluating, and securing ultra-low-power embedded and batteryless devices. I approach resource-constrained systems through a full-stack lens, combining hardware/software co-design, bare-metal firmware development, system-level benchmarking, and performance-energy tradeoff analysis. My research investigates practical security challenges in deployed embedded platforms, including side-channel vulnerabilities, code-reuse threats, trusted execution environments, secure checkpointing, and randomness under intermittent operation.

EDUCATION

PhD, Computer Science

Virginia Tech

2023 - Present

Advisor: Dr. Matthew Hicks

MS, Electrical and Computer Engineering

Virginia Tech

2021 - 2023

GPA: 3.8/4.0

BS, Electronics and Telecommunication Engineering

Dwarkadas J. Sangvhi College of Engineering

2017 - 2021

GPA: 7.8/10.0

PROFESSIONAL EXPERIENCE

Virginia Tech | Graduate Research Assistant

2022 - Present

- Explore the vulnerability space of ultra-low size, weight and power microcontroller trusted execution environments.
- Design and evaluate trusted execution and context management techniques in resource-constrained environments.
- Investigate trade offs between security and performance of checkpointing techniques in intermittently powered batteryless devices.
- Evaluate the robustness of random number generators present in intermittently powered batteryless devices.

Arm | Productivity Engineering Intern

Summer 2024

- Improved the Medini threat modelling process of HSDL (Hardware Secure Development Lifecycle) used in 50+ Arm products.
- Integrated support for Architecture and Technology Group security risk assessment and system microarchitecture threat models.
- Contributed to an in-house microarchitecture vulnerability pen testing tool based on functional verification techniques.

NTPC | Software Engineering Intern

Winter 2020

- Automated the file accessing system and integrated it with the a calendar widget using Javascript.
- Built the database model and front-end design for the company website using ASP.NET MVC and SQL server.

TECHNICAL SKILLS

- Embedded Systems • Intermittent Computation • Hardware Security
- Hardware/software co-design • Threat Modeling & Risk Assessment
- TEE: Texas Instruments (TI) MSP IP Encapsulation, Arm TrustZone
- TI CCS, TI DSS, MPLAB X, Atmel Studio, Keil uVision, mspdebug, GDB
- AMD Vivado Design Suite, Cadence Genus, GCC, ANSYS Medini
- C, Assembly (ARM, MSP430), Verilog, Python, Javascript

SERVICE

Guest Lecturer | CS 4264: Principles of Computer Security

Fall 2025

PUBLICATIONS	μRNG: A Framework for Assessing Randomness in Intermittent Computing Devices. <u>Prakhar Sah</u> and Matthew Hicks. Annual International Symposium on Computer Architecture (ISCA). June 2026.	
	Sheriff: Arresting the Confused Deputy Undermining Ultra-low SWaP Device Trusted Execution Environments. <u>Prakhar Sah</u> and Matthew Hicks. ACM Asia Conference on Computer and Communications Security (AsiaCCS). June 2026.	
	RIPencapsulation: Defeating IP Encapsulation on TI MSP Devices. <u>Prakhar Sah</u> and Matthew Hicks. USENIX WOOT Conference on Offensive Technologies (WOOT). August 2024.	
	Hitchhiker's Guide to Secure Checkpointing on Energy-Harvesting Systems. <u>Prakhar Sah</u> and Matthew Hicks. International Workshop on Energy Harvesting & Energy-Neutral Sensing Systems (ENSsys). November 2023.	
RESEARCH PROJECTS	A Framework for Evaluating Random Number Generators (RNGs) in Intermittent Computing Devices: A systematic framework to evaluate and qualify RNGs under intermittent operation and environmental influence. I apply this framework to 189 commercial devices as well as academic prototypes (SRAM- and RO-based). My analysis finds that 132 devices lack on-chip RNGs, 27 employ RNGs that are insecure, and only 23 provide trustworthy entropy sources. Outcome: One conference paper (ISCA '26)	
	Secure Enclave for Confused Deputy (CD) Threat on ULSWaP Devices: Hardware/software co-design for secure storage and context management in ULSWaP devices, enabling a hardware-enforced secure memory region for execution and secret storage, supporting secure introspection. Prevents CD attacks with minimal overhead (<1% hardware, <3% runtime), validated in hardware and simulation. Outcome: One conference paper (AsiaCCS '26)	
	Interrupt-based Side-Channel Attacks against Commercial TEEs: Texas Instruments' MSP IP Encapsulation aims to provide confidentiality of data stored inside the IPE memory zone; this includes proprietary code and keys. I devised exploits that break this guarantee by leveraging two fundamental drawbacks in IPE design: residual state on context switches and lack of call site verification. Outcome: One conference paper (USENIX WOOT '24)	
	A Survey of Secure Checkpointing Techniques in Energy-Harvesting Devices: A qualitative comparison of threat models, security guarantees, and resource use in secure checkpointing techniques for energy-harvesting devices. The study finds most techniques overlook adversaries with physical access, common in real-world IoT. It proposes a more realistic, practical, and energy-aware adversarial model. Outcome: One workshop paper (ENSsys '23)	
OTHER PROJECTS	Energy Harvesting and Intermittent Computing: Designed my own checkpoint management system trading off resource availability and freshness across power cycles and evaluated forward progress of embedded crypto benchmarks on recorded energy harvesting power traces.	
	Implementation and Evaluation of a Polling Server on ATmega2560: Designed a polling server, evaluated response times for aperiodic & sporadic tasks.	
	Implementation and Mitigation of Spectre Attacks on RISC-V Microarchitecture: Replicated Spectre PHT & RSB on RISC-V BOOM core and explored mitigation strategies.	
	Object Detecting Autonomous Water Vehicle: Designed robot navigation with SLAM, established remote encoder access, interfaced sensors and actuators with the MCU.	
PRESENTATION & TALKS	Sheriff: Arresting the Confused Deputy Undermining Ultra-low SWaP Device Trusted Execution Environments. (AsiaCCS)	2026
	RIPencapsulation: Defeating IP Encapsulation on TI MSP Devices. (WOOT)	2024
	Hitchhiker's Guide to Secure Checkpointing on Energy-Harvesting Systems. (ENSsys)	2023